Stare, Then Look Away: The Science of Afterimages

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Here is a trick you can try. Stare at a brightly colored picture for 30 to 60 seconds, then look at a plain white surface. You will see a ghostly copy of the picture, the same shape and size, but in different colors. This is called an **afterimage**. A science experiment from Science Buddies explains why it happens.

Your eyes see color using special cells called **cone cells**. They sit in a tiny spot at the back of the eye called the **fovea**, which is only about 1 square millimeter in size. There are three types of **cone cells**, loosely called red, green and blue, based on the kind of light they respond to best. When you look at something red, your red cones send a message to your brain that says the object is red.

When red, green and blue light are added together, they make white light. So when you look at something white, all three types of cones are working at once.

But **cone cells** can get tired. If you stare at a red circle for a long time, your red cones become **fatigued** and stop responding for a moment. Then, when you look at a white surface, only your green and blue cones send messages to your brain. So you see a blue-green circle instead.

After several seconds, the red cones recover and the **afterimage** fades away.

You can even predict an **afterimage**'s color. It is the **complementary color** of the original object. **Complementary colors** are two colors of light that make white when combined, such as red and cyan, green and magenta, or blue and yellow. The project invites you to test different colors and see how your own **cone cells** respond.

Word Treasure

afterimage -- A picture you keep seeing for a moment after you look away from something.
cone cells -- Cells in the eye that detect color and fine detail.
fovea -- A tiny spot at the back of the eye that gives the sharpest vision.
fatigued -- Very tired and temporarily unable to work well.
complementary color -- A color that combines with another color of light to make white.

Background you need

Light enters the eye and lands on the retina at the back, where special cells turn it into signals for the brain.

Cone cells handle color, and each of the three types responds best to red, green or blue light.

When red, green and blue light are mixed together, they make white light, which is different from mixing paints.

Cells that work hard for a long time can become tired, and this is what creates the ghostly colors of an afterimage.

Break it down

LEAD: Staring at a color and looking away creates an afterimage
KEY IDEA: Cone cells in the fovea detect color
+- TYPES: Red, green and blue cones
+- MIXING: Red, green and blue light add up to white
EXPLANATION: Tired cones stop responding, so the brain gets incomplete color signals
EXAMPLE: A red circle leaves a blue-green afterimage
RULE: The afterimage is the complementary color of the original
ACTIVITY: Test different colors to see how your cones respond

Why it matters

The experiment shows that what we see depends on how our eye cells and brain work together, not just on the object itself. It is an easy way to explore human biology at home.

Test what you remember

1. What should you do to see an afterimage?

- (A) Stare at a colored picture, then look at a white surface
- (B) Spin around quickly
- (C) Close your eyes for an hour
- (D) Look directly at the Sun

2. Where are cone cells found?

- (A) In the ear
- (B) In the nose
- (C) On the skin
- (D) In the fovea at the back of the eye

3. How many types of cone cells are there?

- (A) Three
- (B) Two
- (C) Ten
- (D) One

4. What do red, green and blue light make when added together?

- (A) White
- (B) Black
- (C) Purple
- (D) Brown

5. After staring at a red circle, what color is the afterimage?

- (A) Yellow
- (B) Red
- (C) Pink
- (D) Blue-green

6. Why does the afterimage fade after several seconds?

- (A) The picture disappears
- (B) The eye closes
- (C) The light gets brighter
- (D) The red cones recover

Answer key (try the quiz first!): 1: A 2: D 3: A 4: A 5: D 6: D

Questions to think about

MY THOUGHTS (at least 20 words):